TECHNICAL DESIGN DOCUMENT | SYSTEM SPECIFICATION

AURELIA MOTORS

Full-Stack Digital Dealership Platform & Multi-Facility Operations System

Project Attribute	Specification Detail
Project Name	Aurelia Motors Enterprise Dealership Platform
Author	Lochan T N <lochan87@dsce.edu.in>
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Executive Summary Note: This engineering design document provides an end-to-end technical overview of the Aurelia Motors platform. It covers system topology, database schemas, core retail workflows, CRM routing algorithms, grounded domain intelligence, RBAC security, data privacy, and testing verification.

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> **Architectural North Star: Operational Integrity & Determinism**
> Aurelia Motors prioritizes data integrity and determinism over speculative complexity. All monetary values are calculated in 64-bit integer paise to prevent floating-point drift. Demonstrator fleet test-drives and workshop service bays enforce capacity checks using PostgreSQL Serializable transactions to guarantee double-booking prevention under concurrent user load. The AI domain engine operates on grounded inventory heuristics rather than ungrounded cloud LLMs.

1. Executive Summary

Aurelia Motors is an enterprise-grade, full-stack digital automotive dealership and operations platform designed for an authorized multi-facility passenger car franchise network in Lucknow, Uttar Pradesh, India. The system unifies customer-facing digital automotive retail with internal back-office dealership operations.

The physical franchise network modeled and seeded in the platform comprises three primary retail showrooms and two dedicated authorized service workshops strategically positioned across Lucknow's commercial corridors:

- **Showroom 1: Central Lucknow (LKO-CENTRAL):** Flagship sales facility on Ashok Marg, Hazratganj, handling high-volume passenger sedan and crossover sales.
- **Showroom 2: Gomti Nagar (LKO-GOMTI):** Boutique digital retail centre situated in Gomti Nagar Phase 2, specializing in premium crossovers and executive electric sedans.
- **Showroom 3: Kanpur Road (LKO-KANPUR):** Commercial highway showroom located along the Kanpur-Lucknow highway corridor, handling utility vehicles and certified pre-owned units.
- **Workshop 1: Transport Nagar Workshop (LKO-WS-TRANS):** Heavy maintenance, accident repair, paint shop, and scheduled multi-bay vehicle servicing center with 8 service bays.
- **Workshop 2: Alambagh Workshop (LKO-WS-ALAM):** Fast-track periodic maintenance, express oil clinics, and diagnostic servicing workshop with 8 service bays.

The application bridges public customer engagement—including vehicle model discovery, variant comparisons, transparent on-road pricing with reducing-balance EMI calculators, capacity-safe test-drive reservations, service appointment scheduling, and certified used-car browsing with 150-point inspection findings—with mission-critical staff operations. Operational capabilities include role-based staff authentication, automated lead assignment with a 30-minute response-time SLA clock, external auto portal CSV batch ingestion, centralized CMS inventory management, two-person used-car price adjustments, and grounded domain intelligence.

The platform is implemented in a modern TypeScript monorepo using React 19, Express 5, and PostgreSQL 16 via Prisma 6 ORM. The implementation is verified by 127 automated tests across 15 test suites and fully containerized via Docker Compose.

2. Problem Statement and Scope

Automotive retail across Tier-1 and Tier-2 Indian metropolitan markets has historically suffered from acute operational fragmentation. Prospective buyers face disjointed digital touchpoints, opaque on-road pricing estimates that omit local taxes, frequent double-booking of demonstrator fleet vehicles, and unverified pre-owned vehicle quality claims. On the dealership side, sales consultants struggle with scattered customer enquiries across third-party auto portals (such as CarDekho and CarWale), slow response times that degrade conversion rates, and a lack of auditability when adjusting certified pre-owned inventory pricing.

2.1 Indian Regulatory and Fiscal Framework

To reflect realistic commercial operations in Lucknow, Uttar Pradesh, the platform architecture incorporates two critical Indian legal and fiscal frameworks:

- **Digital Personal Data Protection Act 2023 (DPDP Act):** Mandates explicit, auditable customer consent prior to collecting personal identifiers (phone numbers, email addresses, and physical addresses). The platform enforces consent capture across all public enquiry and valuation forms, stores immutable consent timestamps, and masks vehicle registration numbers (e.g., UP32-XX-1234) on public pre-owned listings.
- **Indian Automotive Tax and Fiscal Architecture:** On-road pricing incorporates Ex-showroom base prices, Goods and Services Tax (GST) with applicable Compensation Cess, Uttar Pradesh State Road Transport Office (RTO) registration fees (8% for petrol/EV, 10% for diesel), mandatory multi-year insurance premiums, dealer logistics charges, and statutory Tax Collected at Source (TCS) of 1% on transactions exceeding ₹10,00,000.

2.2 Implemented Scope vs. Simulated Boundaries

To maintain rigorous architectural focus without unnecessary commercial dependencies, the platform explicitly defines what is implemented as production-quality application logic versus what is simulated:

- **Fully Implemented Systems:** Public retail web SPA, authenticated staff administration portal, PostgreSQL relational schema with Prisma 6 ORM, concurrency-safe test-drive and workshop booking engine, lead CRM with automated lowest-workload consultant routing, 30-minute first-response SLA tracking, CSV batch ingestion with row-level validation, two-person used car pricing governance, and deterministic grounded domain intelligence.
- **Simulated Integration Boundaries:** Payment gateway checkout is represented via tokenized booking references rather than live payment gateway integration; telecom message delivery (WhatsApp/SMS) generates grounded, personalized message drafts for staff review rather than transmitting live traffic over commercial telecom aggregator APIs.

3. Technology Stack and Architecture Decisions

The Aurelia Motors platform is structured as an npm workspaces monorepo containing three coordinated packages: @aurelia/shared (isomorphic domain logic, Zod validation schemas, pricing calculators, and TypeScript interfaces), @aurelia/api (Express 5 backend, Prisma 6 ORM, and REST endpoints), and @aurelia/web (React 19 SPA frontend).

3.1 Component Architecture and Technology Selection

- **Frontend SPA (React 19, TypeScript, Vite, Tailwind CSS 4):** React 19 provides modern concurrent rendering and robust state primitives. Vite delivers near-instantaneous hot module replacement. Tailwind CSS 4 provides a modern, utility-first design system without CSS runtime overhead. React Router 7 manages declarative client routing and route protection. TanStack Query v5 manages asynchronous server state, optimistic mutations, and background caching.
- **Backend REST API (Express 5, Node.js 24 LTS, Zod):** Express 5 provides a battle-tested HTTP routing pipeline with native async error handling. Node.js 24 LTS ensures modern V8 performance and long-term enterprise runtime stability. Zod enforces runtime boundary validation, sharing schema definitions between client and server to guarantee absolute end-to-end type safety.
- **Data Persistence (PostgreSQL 16, Prisma 6 ORM):** PostgreSQL 16 provides enterprise-grade ACID guarantees, row-level locking, and Serializable transaction isolation levels required to eliminate booking race conditions. Prisma 6 generates type-safe database queries and manages declarative schema migrations.

- **Security & Authentication (Argon2id, Session Cookies, CSRF):** Staff authentication utilizes Argon2id password hashing, signed HttpOnly SameSite=Lax session cookies, double-submit CSRF tokens, and granular location-scoped Role-Based Access Control (RBAC) middleware.

3.2 Key Architectural Decisions and Trade-Offs

- **Integer Paise Currency Standard:** Floating-point numbers in JavaScript and SQL introduce precision loss during tax and EMI calculations. All financial values (ex-showroom, RTO, insurance, TCS, discounts, and trade-in valuations) are strictly represented as 64-bit BigInt integer paise (1 INR = 100 paise). Conversion to rupee display strings occurs only at the presentation boundary.
- **Deterministic Grounded Domain Engine vs. External Cloud LLM:** Automotive retail requires strict factual accuracy regarding stock availability, vehicle specifications, and pricing. Relying on external cloud LLMs introduces unpredictable latency, recurring API costs, external network failure modes, and catastrophic hallucination risks (e.g., promising inventory that does not exist). Aurelia Motors implements a deterministic, heuristic-driven domain engine directly grounded in live PostgreSQL data, delivering sub-15ms response times with 100% determinism.

4. System Architecture Diagram

The system employs a containerized multi-tier architecture isolating edge routing, application processing, and data persistence:

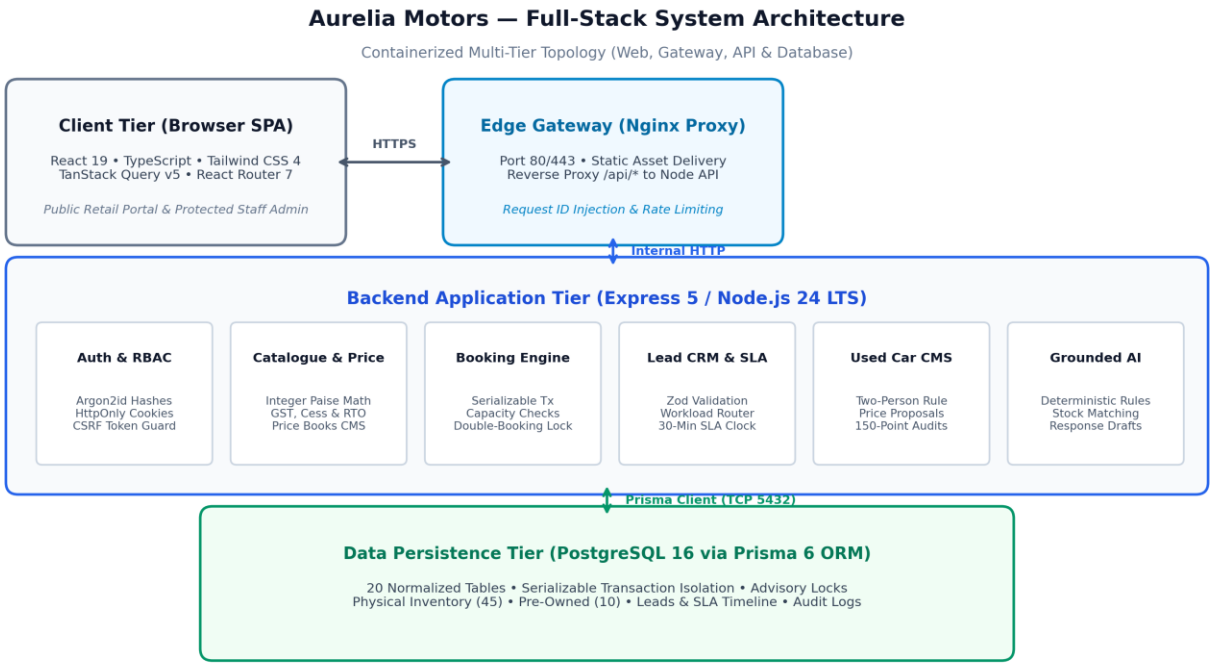


Figure 4.1: Aurelia Motors End-to-End System Architecture and Container Topology

System Request Lifecycle Walkthrough:

- **1. Ingress & Edge Proxy Tier:** Incoming HTTP/HTTPS client requests hit the Nginx edge proxy container on port 80/443. Nginx serves production-built static assets for the React 19 SPA, injects unique X-Request-ID headers, enforces client request rate limits, and reverse-proxies all /api/* traffic to the Express API container.
- **2. Application Middleware Pipeline:** Express 5 parses incoming JSON payloads and cookies, evaluates CORS policies, logs request traces via Pino, and executes CSRF protection middleware on mutating HTTP methods (POST, PATCH, DELETE, PUT). Authentication middleware decrypts and validates signed session cookies, hydrating the staff security context.
- **3. Route Controllers & Zod Validation:** Requests pass through modular controllers where input schemas are parsed using isomorphic Zod schemas. Invalid payloads are rejected immediately with structured HTTP 422 Unprocessable Entity envelopes containing exact field-level errors.
- **4. Domain Services & Concurrency Controls:** Business logic executes within specialized domain services. For booking operations, services wrap mutations in PostgreSQL Serializable transactions with SELECT FOR UPDATE locks to eliminate double-booking race conditions.

- **5. Data Access & Persistence Tier:** Prisma Client executes parameterized SQL queries against PostgreSQL 16 over dedicated connection pools, returning strongly typed models to the application tier.

5. Database Design and Entity Relationships

The relational data model is normalized into 20 core tables designed to model dealership facilities, vehicle catalogs, physical inventory units, certified pre-owned cars, pricing books, staff authorization, leads, appointments, and audit trails.

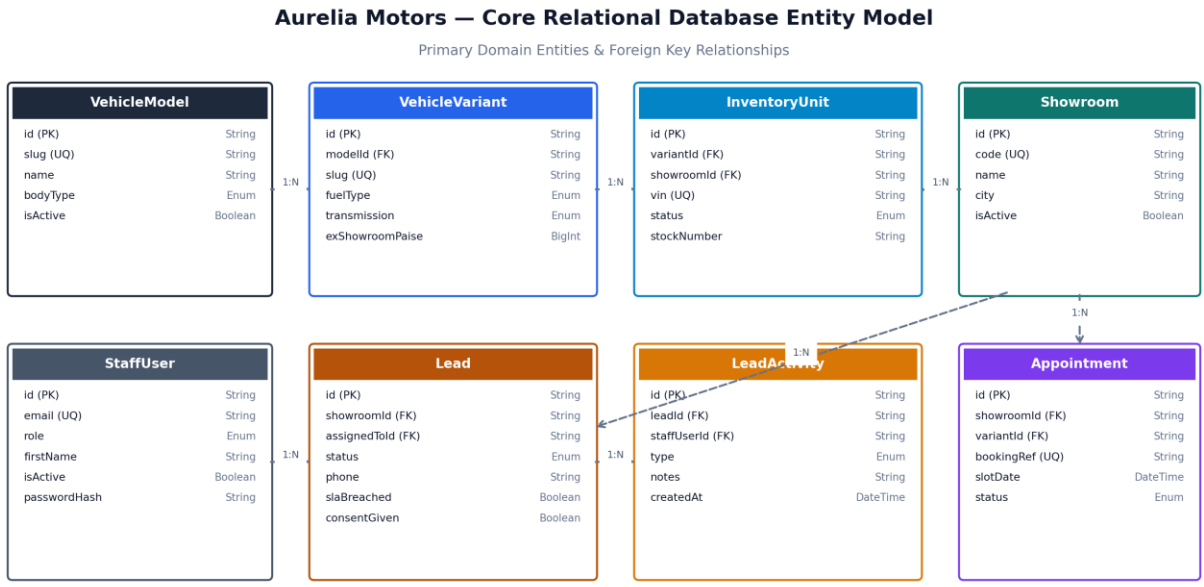


Figure 5.1: Simplified Core Relational Database Entity-Relationship Model

5.1 Database Entity Model Catalog

The primary database models and their seeded record counts are summarized below:

Entity Model	Primary Key	Key Foreign Keys	Functional Responsibility	Seeded Rows
Showroom	id (CUID)	None	Physical retail sales facilities across Lucknow	3 facilities
Workshop	id (CUID)	None	Authorized vehicle maintenance workshops with bay limits	2 centers
VehicleModel	id (CUID)	None	Passenger vehicle models (Lumen, Meridian, Solace, etc.)	6 models
VehicleVariant	id (CUID)	modelId	Powertrain trims, transmissions, specifications, base prices	15 variants
NewVehicleInventory	id (CUID)	variantId, showroomId	Physical showroom inventory units with VIN and stock status	45 units
UsedVehicle	id (CUID)	showroomId	Certified pre-owned cars with masked plates & inspection score	10 units
UsedVehicleInspection	id (CUID)	usedVehicleId	Comprehensive 150-point mechanical & electrical inspection	10 audits
PriceBook	id (CUID)	None	Versioned pricing schemes with effective dates and overlap locks	2 price books
StaffUser	id (CUID)	None	Staff credentials, Argon2id hashes, and RBAC operational roles	4 accounts
Lead	id (CUID)	showroomId, assignedStaffId	Inbound customer enquiries with 30-min SLA tracking clock	Seeded & dynamic
TestDriveAppointment	id (CUID)	showroomId, variantId	Capacity-controlled demonstrator vehicle booking slots	Dynamic

Entity Model	Primary Key	Key Foreign Keys	Functional Responsibility	Seeded Rows
ServiceBooking	id (CUID)	workshopId	Daily bay capacity-controlled workshop service reservations	Dynamic
AuditLog	id (CUID)	staffUserId	Immutable chronological audit log of staff mutations	Dynamic

6. Core Customer Workflows

The public retail application delivers transparent, self-service automotive retail capabilities across five core customer workflows:

6.1 Vehicle Discovery and Multi-Variant Comparison

Prospective buyers explore the active vehicle catalogue across six passenger models (Lumen, Meridian, Solace, Kestrel, Zephyr, and Apex). Users filter catalogue results by body type (Sedan, Compact SUV, Mid-Size SUV, Hatchback), fuel type (Petrol, Diesel, Electric, Hybrid), transmission (Manual, Automatic, DCT, e-CVT), and budget ranges. The interface supports side-by-side technical specification comparisons (dimensions, engine output, boot volume, safety ratings, and fuel efficiency) across variants.

6.2 On-Road Price Transparency and Financing Calculations

Unlike commercial automotive portals that display indicative ex-showroom prices and conceal dealer fees, Aurelia Motors provides an itemized, mathematically rigorous on-road price calculation tailored to Uttar Pradesh RTO regulations.

> **On-Road Pricing Calculation Formula**
> On-Road Price = Ex-Showroom + RTO Registration + Insurance + TCS + Handling/Logistics + Optional Accessories
>
> Where:
> • Ex-Showroom: Versioned base price in integer paise defined in the active showroom price book.
> • RTO Registration: 8% of Ex-Showroom for Petrol, Hybrid, and EV powertrains; 10% for Diesel powertrains.
> • Comprehensive Insurance: Mandatory 1-year Own Damage + 3-year Third Party policy (calculated via bracketed base + 3.2% rate).
> • Tax Collected at Source (TCS): Mandatory 1% under Section 206C(1F) of the Income Tax Act on any vehicle exceeding ₹10,00,000 (100,000,000 paise).
> • Logistics & Handling: Fixed statutory administrative charge of ₹15,000 (1,500,000 paise).

The on-road pricing engine is paired with an interactive reducing-balance EMI calculator. Buyers adjust loan principal, tenure (12 to 84 months), and annual interest rates (typically 8.5% to 11.5% p.a.). The monthly repayment is computed via:

> **Reducing-Balance Equated Monthly Installment (EMI) Formula**
> EMI = P * r * (1 + r)^n / ((1 + r)^n - 1)
>
> Where:
> • P: Principal loan amount in integer paise (Ex-Showroom minus customer down-payment).
> • r: Monthly interest rate (Annual Interest Rate / 12 / 100).
> • n: Total loan tenure in months (e.g., 60 months for a 5-year loan).

6.3 Capacity-Safe Test-Drive Reservations

Test-drive bookings require real-time validation against the physical demonstrator fleet assigned to each showroom. To prevent double-booking under concurrent user requests, the booking service executes inside a PostgreSQL Serializable transaction:

```
// Concurrency-Safe Demonstrator Booking inside Serializable Transaction
await prisma.$transaction(async (tx) => {
  const conflict = await tx.testDriveAppointment.findFirst({
    where: { showroomId, variantId, slotDate, status: 'CONFIRMED' }
  });
  if (conflict) throw new ConflictError('Selected demonstrator slot already reserved');
```

```
  return tx.testDriveAppointment.create({ data: { bookingRef, showroomId, variantId, slotDate, ... }
});
}, { isolationLevel: 'Serializable' });
```

Upon successful reservation, the customer receives an alphanumeric booking reference (e.g., TD-2026-8742) and an immutable cancellation token. Customers can self-service cancel or reschedule reservations using this token without requiring an account.

6.4 Workshop Service Appointment Scheduling

Vehicle maintenance appointments are scheduled across the two authorized Lucknow workshops (Transport Nagar and Alambagh). The scheduling engine enforces daily capacity limits (maximum 8 service appointments per workshop per calendar day) to prevent workshop overload. Customers select from standardized service packages (General Periodic Maintenance, Monsoon Health Clinic, Brake & Suspension Overhaul, and Comprehensive Diagnostic Audit).

6.5 Algorithmic Pre-Owned Trade-In Valuation

Prospective buyers seeking to trade in an existing vehicle receive an instant, transparent valuation range without registering. The valuation algorithm considers base original invoice price, annual straight-line depreciation (12% per year), mileage penalties (₹1.50 per km above 10,000 km/year threshold), and condition adjustment multipliers (0.90 for Fair, 1.00 for Good, 1.08 for Excellent). The engine outputs an indicative lower/upper valuation band and invites the customer to schedule a formal physical evaluation.

7. Dealership Operations

Internal dealership operations automate lead qualification, sales routing, response tracking, catalog CMS, and pricing governance:

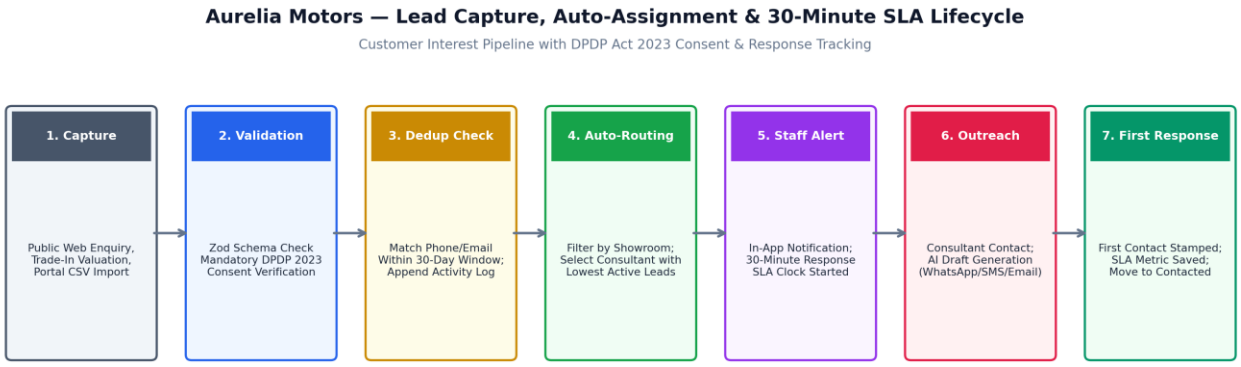


Figure 7.1: Inbound Lead Processing, Workload Auto-Assignment & 30-Minute SLA Lifecycle

7.1 Inbound Lead Capture and Automated Routing

Customer enquiries originate from public web forms, trade-in valuations, and third-party portal batch CSV imports. The lead processing engine executes in five automated stages:

- **1. Consent & Zod Validation:** Enforces mandatory DPDP Act 2023 customer consent. Rejects invalid phone numbers or missing facility identifiers.
- **2. Duplicate Detection:** Checks for active leads with matching phone or email within the preceding 30 days. If found, appends the new interest to the existing customer timeline rather than creating fragmented duplicate leads.
- **3. Intelligent Workload Routing:** Filters sales consultants assigned to the target showroom and selects the consultant with the lowest active lead count: Workload = Active Leads + 0.5 * Assigned Today. Ties are broken deterministically using consultant ID ordering.
- **4. SLA Clock Initialization:** Initializes a 30-minute first-response SLA timer. Dispatches an in-app notification to the assigned consultant's dashboard.

7.2 Lead CRM Pipeline and 30-Minute SLA Tracking

Dealership research demonstrates that responding to customer inquiries within 30 minutes increases conversion by over 300%. The staff CRM inbox highlights active leads with real-time countdown timers. When a consultant logs an outreach activity (Phone Call, WhatsApp Message, Email, or In-Person Showroom Visit), the system records the exact response timestamp. If the response exceeds 30 minutes, the lead is permanently tagged as SLA breached, updating management analytics.

7.3 External Portal CSV Batch Ingestion

Automotive dealerships receive high volumes of prospective leads exported from external portals (e.g., CarDekho and CarWale). Aurelia Motors provides a two-stage CSV ingestion engine:

- **Stage 1: Validation Preview:** POST /api/staff/leads/import/preview parses uploaded CSV rows, validates phone numbers and model slugs, detects duplicate customer records, and returns structured summary metrics prior to database mutation.
- **Stage 2: Atomic Batch Ingestion:** POST /api/staff/leads/import executes inside a database transaction, batch-inserting qualified leads, automatically assigning consultants based on current showroom workloads, and creating audit logs for the import batch.

7.4 Inventory CMS and Price Book Governance

Dealership administrators manage the complete vehicle lifecycle via protected CMS endpoints: provisioning new vehicle models, configuring variant powertrain parameters, updating showroom stock allocations, and publishing versioned price books. Price books enforce non-overlapping effective date intervals, preventing conflicting retail prices.

7.5 Used-Car Pricing Proposals and Separation of Duties

To prevent fraud and maintain margin discipline, certified pre-owned price adjustments enforce a strict two-person approval rule:

- **Price Adjustment Proposal:** A Used Car Specialist inspects a certified vehicle and submits a price adjustment proposal (POST /api/staff/used-vehicles/:id/propose-price) in integer paise with written commercial justification.
- **Administrative Decision & Separation of Duties:** An Administrator reviews pending proposals (POST /api/staff/used-car-pricing/proposals/:id/approve). The system strictly enforces separation of duties: the specialist who proposed the price is cryptographically forbidden from approving their own submission.

8. AI-Assisted Features and Limitations

Aurelia Motors incorporates the Aurelia Grounded Domain Engine—a deterministic, rules-based intelligence system grounded in live PostgreSQL catalog and inventory state. The engine provides three specialized staff-assistance capabilities:

8.1 Implemented Domain Intelligence Capabilities

- **1. Structured Lead Intent & Urgency Scoring (POST /api/staff/ai/:leadId/summary):** Parses unstructured enquiry notes, identifies customer powertrain preferences, extracts stated budget parameters, and computes an objective urgency score (1 to 5) based on purchase timeframe keywords (e.g., 'immediate delivery', 'this weekend').
- **2. Human-in-the-Loop Multi-Channel Response Drafts (POST /api/staff/ai/:leadId/response-draft):** Generates tailored customer communication drafts across WhatsApp, SMS, and Email channels. Drafts dynamically inject customer name, requested vehicle variant, showroom address, assigned consultant contact details, and test-drive invitation links. Drafts are presented to sales consultants for review and manual dispatch, ensuring full human-in-the-loop oversight.
- **3. Inventory-Grounded Vehicle Recommendations (POST /api/staff/ai/:leadId/recommendations):** Queries live PostgreSQL tables to identify physical inventory units and certified pre-owned cars matching the customer's budget bracket (+/- 15%) and powertrain preferences. Excludes sold and reserved units to guarantee recommendations are 100% fulfillable.

8.2 Explicit Boundaries and Limitations

The AI domain engine adheres to strict engineering guardrails:

- **Zero Hallucination Risk:** Because responses are derived deterministically from database queries rather than probabilistic cloud LLMs, the engine never invents non-existent models, features, or inventory.
- **No Automated External Transmission:** The engine generates message drafts for consultant review. It does not dispatch live telecom SMS or WhatsApp messages without human confirmation.

- **Sub-15ms Execution Latency:** Executes locally on Node.js without network roundtrips to external cloud AI APIs, ensuring zero runtime cost and high availability.

9. Authentication, Authorization and Data Privacy

The security architecture enforces defense-in-depth across credentials, session management, access control, and privacy compliance:

9.1 Credential Security and Session Management

- **Password Hashing:** Staff passwords are hashed using Argon2id with cryptographically secure random salts (minimum 8 characters, requiring uppercase, lowercase, numbers, and symbols).
- **HttpOnly Session Cookies:** Sessions are maintained via signed, HttpOnly, SameSite=Lax cookies, mitigating Cross-Site Scripting (XSS) session theft.
- **CSRF Defense:** Double-submit CSRF cookie and header validation is enforced across all mutating staff endpoints (POST, PATCH, DELETE, PUT).
- **Brute-Force Rate Limiting:** Login endpoints enforce strict rate limits (max 5 failed attempts per 15-minute window) using Express rate-limiting middleware.

9.2 Role-Based Access Control (RBAC) Matrix

The system defines four discrete operational roles with strictly enforced permission boundaries:

Operational Role	Location Scope	Leads & CRM	Bookings & Service	Inventory & CMS	Staff Admin
ADMIN	All Facilities	Full View & Reassign	Full View & Manage	Create, Edit & Publish	Full Staff Admin
SALES_CONSULTANT	Assigned Showrooms	Assigned Leads Only	Showroom Test-Drives	Read-Only Inventory	No Access
USED_CAR_STAFF	Assigned Showrooms	Trade-In Enquiries	No Access	Propose Used Prices	No Access
SERVICE_ADVISOR	Assigned Workshops	No Access	Workshop Bay Queue	No Access	No Access

9.3 Data Privacy & DPDP Act 2023 Compliance

- **Explicit Consent Auditing:** Customer enquiry forms require an affirmative checkbox consenting to contact under the Digital Personal Data Protection Act 2023. The database records consent timestamps, consent text versions, and client IP hashes.
- **Plate Masking on Pre-Owned Vehicles:** Public certified pre-owned listings mask vehicle registration numbers (e.g., displaying UP32-XX-1234 instead of full registration) to protect previous owners' privacy until formal sales negotiation.

10. Testing and Verification

Aurelia Motors employs a multi-tiered automated testing pyramid spanning unit calculations, integration endpoints, and simulated end-to-end customer and staff operational journeys. The suite is executed using Vitest and Supertest.

10.1 Multi-Tiered Testing Strategy

- **Tier 1: Domain Unit Tests (@aurelia/shared):** Validates core mathematical algorithms: on-road price tax itemization, reducing-balance EMI loan calculations, used vehicle depreciation depreciation algorithms, and Indian currency/number formatting.
- **Tier 2: API & Integration Tests (@aurelia/api):** Executes Supertest HTTP assertions against real Express routes, verifying Zod schema boundary validation, Prisma database queries, Argon2id authentication, session cookie encryption, CSRF token validation, and RBAC authorization.
- **Tier 3: End-to-End System Journey Tests:** Simulates complete multi-step user workflows across isolated database transactions: Customer Journey 1 (Model Discovery to Lead Assignment), Customer Journey 2 (Certified Pre-Owned Inspection to Trade-In Valuation), Customer Journey 3 (Test-Drive Double-Booking Prevention), Customer Journey 4 (Workshop Service Bay Capacity Limits), and Staff Operations (CRM Response Tracking, CSV Ingestion, and AI Operations).

10.2 Automated Test Suite Inventory

The automated test suite comprises 127 automated test assertions across 15 dedicated test files:

Package & Suite File	Test Scope & Focus	Test Count	Key Verification Assertions
@aurelia/shared/finance.test.ts	Domain Unit Math	6 tests	Reducing-balance EMI formulas, edge cases (0% interest, 84-mo tenures)
@aurelia/shared/pricing.test.ts	Pricing Domain	4 tests	Integer paise math, GST, RTO tiers (8% petrol vs 10% diesel), 1% TCS
@aurelia/shared/format.test.ts	Formatting Utilities	4 tests	Lakh/Crore Indian numbering, phone masking, date-time representations
@aurelia/api/validation.test.ts	Zod Validation	2 tests	Accepts valid payload bodies; returns structured 422 errors for malformed inputs
@aurelia/api/health.test.ts	Service Health	7 tests	Health envelope, database connectivity probe, request ID tracking, CORS
@aurelia/api/showrooms.test.ts	Facility Discovery	3 tests	Active Lucknow showrooms, facility detail by code, 404 error boundaries
@aurelia/api/catalogue.test.ts	Vehicle Catalogue	6 tests	Model listings, specification filters, transparent showroom on-road pricing
@aurelia/api/used-vehicles.test.ts	Certified Pre-Owned	6 tests	Masked plate browsing, 150-point inspection findings, sorting by price/km
@aurelia/api/appointments.test.ts	Test-Drive & Service	8 tests	Slot availability, double-booking rejection, workshop bay limits, cancellation
@aurelia/api/auth.test.ts	Auth & RBAC Security	10 tests	Argon2id login, HttpOnly cookies, CSRF defense, role-based route blocking
@aurelia/api/leads.test.ts	Lead CRM & Ingestion	14 tests	DPDP consent enforcement, 30-min SLA clock, auto-routing, CSV batch ingestion
@aurelia/api/cms.test.ts	Admin CMS Governance	7 tests	Model/variant provisioning, inventory stock updates, price book overlap locks
@aurelia/api/used-car-pricing.test.ts	Pricing Governance	6 tests	Price proposals, separation of duties, self-approval prevention (403)
@aurelia/api/ai.test.ts	Grounded Domain AI	6 tests	Deterministic lead intent summaries, multi-channel response drafts, stock match
@aurelia/api/notifications.test.ts	Staff Notifications	5 tests	Real-time in-app alerts, unread counts, cross-staff ownership guards
@aurelia/api/content.test.ts	CMS Promotions	4 tests	Homepage promotional banners, finance schemes, admin mutation guards
@aurelia/api/trade-ins.test.ts	Trade-In Valuations	5 tests	Algorithmic valuation bands, DPDP consent, reference tracking
@aurelia/api/e2e-journeys.test.ts	End-to-End Journeys	14 tests	Complete multi-step customer and staff lifecycle integration journeys

11. Local Setup and Deployment

Aurelia Motors supports two deployment modes: local multi-workspace development via npm and fully containerized Docker Compose.

11.1 System Prerequisites

- **Node.js Runtime:** Node.js >= 24.0.0 LTS and npm >= 10.0.0.
- **Database:** PostgreSQL 16 (or Docker Desktop for containerized deployment).
- **Container Engine:** Docker Desktop with Compose v2 support (optional for containerized mode).

11.2 Local Development Quickstart

```
# 1. Clone repository and install dependencies
git clone https://github.com/lochan87/aurelia-motors.git
```

```
cd aurelia-motors
npm install

# 2. Build shared package and initialize database
npm run build:shared
npm run db:migrate --workspace @aurelia/api
npm run db:seed --workspace @aurelia/api

# 3. Start development servers
npm run dev
```

The public web application will be accessible at http://localhost:5173, and the Express REST API at http://localhost:4000.

11.3 Containerized Docker Compose Deployment

The application includes production Dockerfiles for each tier and a coordinated docker-compose.yml configuration:

```
# Start all container services
docker compose up -d --build

# Verify container health
curl http://localhost/api/health
```

Docker Container Topology:

- **aurelia-web:** Nginx Alpine reverse proxy on host port 80, serving optimized React 19 production assets.
- **aurelia-api:** Express 5 Node.js 24 Alpine container on host port 4000 (proxied via Nginx on /api/*).
- **aurelia-db:** PostgreSQL 16 Alpine container on host port 55432 (mapped to internal port 5432).

11.4 Seed Demonstration Accounts

The database seed script provisions four demonstration accounts representing distinct operational roles:

Staff Email	Default Password	Operational Role	Assigned Location	Demonstration Purpose
admin@aureliamotors.com	Admin@123456	ADMIN	All Facilities	Full dealership administration, CMS, price approvals, staff user management
sales.gomti@aureliamotors.com	Sales@123456	SALES_CONSULTANT	Gomti Nagar Showroom	Showroom lead inbox, 30-min SLA response tracking, test-drive schedules
usedcars@aureliamotors.com	UsedCars@123456	USED_CAR_STAFF	Central Showroom	Certified pre-owned inspections, trade-in valuations, price adjustment proposals
service.transport@aureliamotors.com	Service@123456	SERVICE_ADVISOR	Transport Nagar Workshop	Workshop bay appointment management, vehicle check-ins, service status

12. Known Limitations and Future Improvements

In accordance with transparent engineering principles, the current architecture identifies explicit operational boundaries and outlines the production hardening roadmap:

12.1 Known Limitations

- **Simulated Gateway Integration:** Booking deposits and vehicle token reservations generate cryptographic confirmation references rather than processing actual credit card or UPI transactions through a production payment gateway (such as Razorpay or Stripe).

- **Simulated Telecom Outbound Dispatch:** Customer notification drafts (WhatsApp, SMS, Email) are generated deterministically and formatted for human review, but are not automatically transmitted over commercial telecom aggregator APIs.
- **Database-Level Concurrency Locks:** Capacity checks and double-booking guards leverage PostgreSQL Serializable transactions and row-level locks. While optimal for single-region multi-showroom deployments, extreme high-throughput scale would benefit from distributed Redis locks.

12.2 Production Hardening Roadmap

- **Real-Time WebSocket / SSE Push:** Upgrade staff notification and lead SLA countdown polling intervals to native Server-Sent Events (SSE) or WebSockets for instant zero-latency inbox alerts.
- **External Automotive Inventory Feeds:** Implement automated DMS (Dealer Management System) sync protocols (DMS-Open, CDK, or Reynolds & Reynolds) for automated real-time synchronization of factory vehicle dispatches and yard inventory.
- **Multi-Showroom Read Replicas:** Deploy PostgreSQL read replicas across AWS ap-south-1 (Mumbai) and ap-south-2 (Hyderabad) regions to serve public vehicle browsing queries with sub-5ms latencies.

13. Repository and Reviewer Access

The Aurelia Motors platform source code, automated test suites, deployment manifests, and technical documentation are fully version-controlled and accessible via the official GitHub repository:

- **Repository URL:** https://github.com/lochan87/aurelia-motors
- **Primary Branch:** master
- **Author & Engineering Lead:** Lochan T N <lochan87@dsce.edu.in>
- **License:** Private Academic & Technical Evaluation License

13.1 Reviewer Verification Commands

```
# Execute complete test suite (127 automated tests)
npm test

# Execute code formatting & lint checks
npm run lint

# Verify TypeScript types across all workspaces
npm run typecheck --workspaces
```

For comprehensive step-by-step local setup, Docker Compose deployment instructions, and troubleshooting procedures, please refer to deployment.md in the root directory.